Day 4: Generation Parameters

Gen-AI + Agentic AI with JavaScript (Gemini Edition) | Phase 1: LLM Foundations

1. Concept

Day 3 me samjha tha ki LLM har step pe agle token ki **probabilities** nikalta hai, phir ek token chunta hai. Aaj dekhenge ki **chunne ka tareeka tum control kaise karte ho.**

Ek example

Prompt: "`Aaj mausam bahut ___`". Model ye probabilities deta hai:

Token	Probability
accha	45%
kharab	20%
garam	15%
thanda	10%
ajeeb	5%
(baaki hazaaron tokens)	5%

Temperature: randomness ka knob

Temperature chunne se pehle probabilities ka **shape** badalta hai.

Temperature	Kya hota hai	Analogy
Low (0 ke paas)	Sabse likely token lagbhag hamesha chuna jata hai. Jawab predictable aur consistent.	Chai wala jo hamesha same order banata hai
High (1 ya upar)	Kam likely tokens ke chance badhte hain. Jawab alag-alag aur creative, par kabhi bhatka hua.	Chai wala jo roz naya experiment karta hai

Use case	Temperature
Data extraction, JSON, classification, code	Low
Brainstorming, story, naam suggestions	Higher

topK: sirf top K options

Sirf **K sabse likely tokens** me se chuno, baaki sab ignore. `topK = 2` ka matlab sirf **accha** aur **kharab** me se chunega.

topP (nucleus sampling): cumulative probability cut

Tokens ko likely se kam likely order me lo, aur tab tak jodo jab tak total **P** tak na pahunch jaye. `topP = 0.8` ka matlab accha (45) + kharab (20) + garam (15) = **80%**, yaani sirf ye 3 candidates.

> **topK vs topP:** topK me options ki *sankhya* fixed hoti hai. topP me options ki sankhya **situation ke hisaab se badalti hai** (jahan model confident hai wahan kam, jahan unsure hai wahan zyada).

maxOutputTokens: hard cap

Ye response ki **maximum length** hai.

> **Sabse common beginner galti:** Log sochte hain ki maxOutputTokens: 50 karne se model chhota aur poora jawab dega. Aisa **nahi** hota. Model ko cap ka pata hi nahi hota, wo bas beech me **kaat diya jata**

hai (truncate). Chhota jawab chahiye toh **prompt me** bolo ("2 lines me batao").

stopSequences

Koi specific text aate hi generation ruk jaye. Jaise `["3."]` daaloge toh list teesre point se pehle ruk jayegi.

Gemini 3 ka important twist

Ye sab concepts har LLM pe lagte hain (OpenAI, Claude, Llama sab), isliye interview ke liye yaad rakho. Lekin **Gemini 3 ke liye practical advice alag hai:**

- Google Gemini 3 models ke liye temperature ko default **1.0** pe hi rakhne ki strongly recommend karta hai, kyunki 1.0 se neeche karne par looping ya reasoning performance me gadbad ho sakti hai.
- Gemini 3 default me dynamic thinking use karta hai. `thinking_level` set na karne par default **high** hota hai, jisse time aur tokens dono zyada lagte hain.
- `thinking_level` aur purana `thinking_budget` ek hi request me saath nahi chalte, warna 400 error aata hai.
- Ek Firebase reference me ye dikha ki Gemini 3.x me kuch sampling parameters (topP/topK jaise) ignore ho sakte hain. Reference ka text jumbled tha, isliye pakka nahi hai. Code me khud dekho ki tumhare model pe kya hota hai.

Takeaway: Gemini 3 pe temperature/topP/topK se tuning karne ki jagah **thinkingLevel, prompt aur structured output** zyada kaam aate hain. Aaj ke experiments seekhne ke liye hain. Production code me Gemini 3 pe temperature low mat karna.

2. Workflow

```
Model ke paas next-token probabilities
        |
        v
topK  -> sirf top K tokens rakho
        |
        v
topP  -> cumulative P tak ke tokens rakho
        |
        v
temperature -> bache hue tokens ka randomness set karo
        |
        v
Ek token chuna -> output me jodo -> repeat
        |
        v
Ruko jab: stop token / stopSequence / maxOutputTokens
```

Experiment ka rule: Ek baar me **ek hi parameter** badlo, nahi toh pata nahi chalega ki farak kisse aaya.

3. Code

Folder: phase-1-foundations/day-04. Setup pehle jaisa (npm init -y, "type": "module", npm i @google/genai dotenv, .env copy).

config.js

```
export const MODEL = "gemini-3.5-flash"; // jo tumhare AI Studio me chal raha ho wo daalo

// Thinking ko chhota rakhte hain taaki experiments tez aur sasta rahe.
// Gemini 3.x  -> thinkingLevel ("low" / "high" ...)
// Gemini 2.5  -> thinkingBudget (jaise 0)  [thinkingLevel yahan error dega]
export const THINKING = { thinkingConfig: { thinkingLevel: "low" } };
```

helpers.js

```
import "dotenv/config";
import { GoogleGenAI } from "@google/genai";
import { MODEL, THINKING } from "./config.js";

const ai = new GoogleGenAI({});

export const sleep = (ms) => new Promise((r) => setTimeout(r, ms));

export async function ask(prompt, extraConfig = {}) {
  return ai.models.generateContent({
    model: MODEL,
    contents: prompt,
    config: { ...THINKING, ...extraConfig },
  });
}
```

temperature.js: same prompt, alag temperature

```
import { ask, sleep } from "./helpers.js";

const prompt = "Suggest one catchy name for a chai cafe. Reply with only the name.";
const temps = [0, 1, 2];

for (const temperature of temps) {
  console.log(`\n=== temperature: ${temperature} ===`);
  for (let i = 1; i <= 3; i++) {
    try {
      const res = await ask(prompt, { temperature });
      console.log(`Run ${i}: ${res.text?.trim()}`);
    } catch (err) {
      console.log(`Run ${i}: ERROR -> ${err.message}`);
    }
    await sleep(1000); // free tier ke liye thoda gap
  }
}
```

Dekhna: Temperature 0 pe 3 runs kitne similar hain? Aur 2 pe kitne alag? (0 ka matlab lagbhag same, 100% guarantee nahi.)

top-p-k.js: topP aur topK

```
import { ask, sleep } from "./helpers.js";

const prompt = "Complete the sentence with one word: The weather today is very";

const settings = [
  { label: "topK=1 (sabse likely hi)", topK: 1 },
  { label: "topK=40", topK: 40 },
  { label: "topP=0.1 (bahut tight)", topP: 0.1 },
  { label: "topP=0.95 (wide)", topP: 0.95 },
];

for (const { label, ...cfg } of settings) {
  console.log(`\n=== ${label} ===`);
  for (let i = 1; i <= 3; i++) {
    try {
      const res = await ask(prompt, { temperature: 1, ...cfg });
      console.log(`Run ${i}: ${res.text?.trim()}`);
    } catch (err) {
      console.log(`Run ${i}: ERROR -> ${err.message}`);
    }
    await sleep(1000);
  }
}
```

Dekhna: Kya topK=1 hamesha same word deta hai? Agar topP/topK ka koi farak nahi dikha, toh ho sakta hai wo parameter tumhare model pe ignore ho raha ho.

max-tokens.js: truncation ka proof

```
import { ask } from "./helpers.js";

const prompt = "Explain how the internet works in detail.";

for (const maxOutputTokens of [30, 1000]) {
  const res = await ask(prompt, { maxOutputTokens });
  const reason = res.candidates?.[0]?.finishReason;
  console.log(`\n=== maxOutputTokens: ${maxOutputTokens} ===`);
  console.log("finishReason:", reason);          // STOP ya MAX_TOKENS
  console.log("Output tokens:", res.usageMetadata?.candidatesTokenCount);
  console.log("Text:", res.text ?? "(khaali)");
}
```

Dekhna: finishReason MAX_TOKENS aaya toh jawab beech me kata gaya. Agar bahut chhota cap rakhne par text **khaali** aaye, toh thinking tokens ne budget kha liya (Gemini ka ek gotcha).

stop-sequence.js

```
import { ask } from "./helpers.js";

const res = await ask("List 5 benefits of learning Node.js as a numbered list.", {
  stopSequences: ["3."], // "3." aate hi ruk jao
});

console.log(res.text);
console.log("finishReason:", res.candidates?.[0]?.finishReason);
```

Total lagbhag 20 calls hongi. Free tier limits ka dhyan rakhna, aur 429 aaye toh thodi der ruk ke dobara chalana.

Homework

1. Chaaron scripts **khud type karke** chalao.
2. temperature.js me ek creative prompt (2-line kavita) aur ek strict prompt (2+2 kitna hota hai, sirf number) try karo, aur dekho temperature ka asar kahan zyada dikhta hai.
3. THINKING ko "high" pe badal ke max-tokens.js dobara chalao aur thoughtsTokenCount dekho.
4. Notes me likho: "Mere model pe topP/topK ka asar dikha ya nahi?"

4. Quick Revision (Quiz Answers)

Question	Answer
Temperature bahut low rakhne se?	Sabse likely token zyada baar chuna jata hai, jawab predictable hote hain
topP = 0.8 pe kitne tokens? (45, 20, 15, 10, 5)	3 tokens (accha, kharab, garam), kyunki 45 + 20 + 15 = 80
topK vs topP?	topK fixed sankhya (K) ke tokens rakhta hai, topP situation ke hisaab se badalti sankhya
maxOutputTokens: 50 pe long answer?	Jawab beech me kat sakta hai, kyunki ye sirf hard limit hai
Gemini 3 ki recommendation?	Temperature default 1.0 pe rakho, aur control ke liye thinkingLevel/prompt use karo

Next: Day 5: System instructions, roles, multi-turn chat